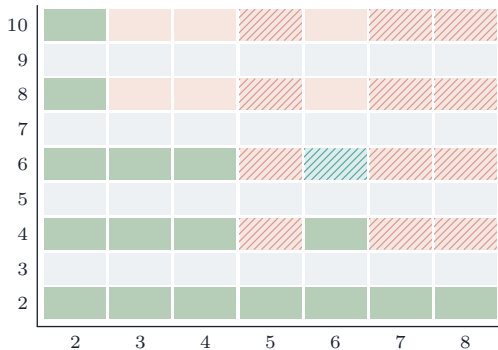


$r = \dim X$  d , the degree of the equations

- proved for every N : the theorem of Lefschetz for $r = 2$, and the very general theorem for $d = 2$, for $d = 3, 4$ with $r \leq 6$ and for $d = 6$ with $r \leq 4$
- proved except for the Weil classes of abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$
- open: Weil classes of abelian r -folds over $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, from dimension 8 on
- open for N large: Weil classes of $\mathbb{Q}(\zeta_e)$, $\varphi(e) \geq 4$, the case $P2_{\text{split}}$ of the paper
- r odd: no Hodge classes of degree r , nothing to prove